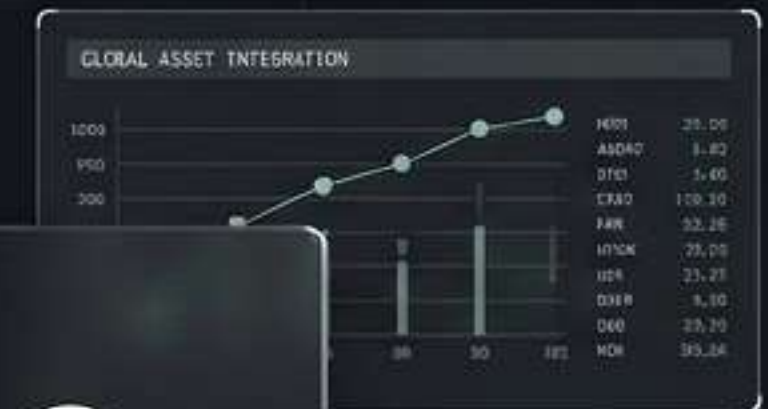
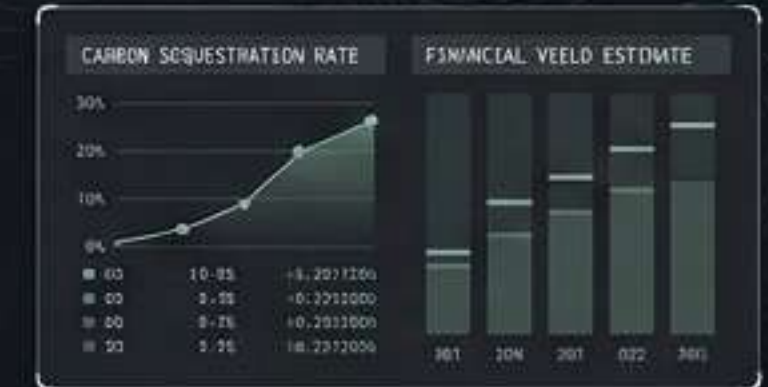


```
SYSTEM BOOT: SUCCESS
MODULE: Planetary Operating System - Architecture
        & Financial Viability
```



PBPE Marketplace

A planetary-scale financial and environmental OS that inverts climate investment from a 'cost' to 'overwhelming profit'.



REQUIRED: \$1.2T - \$1.7T / Year

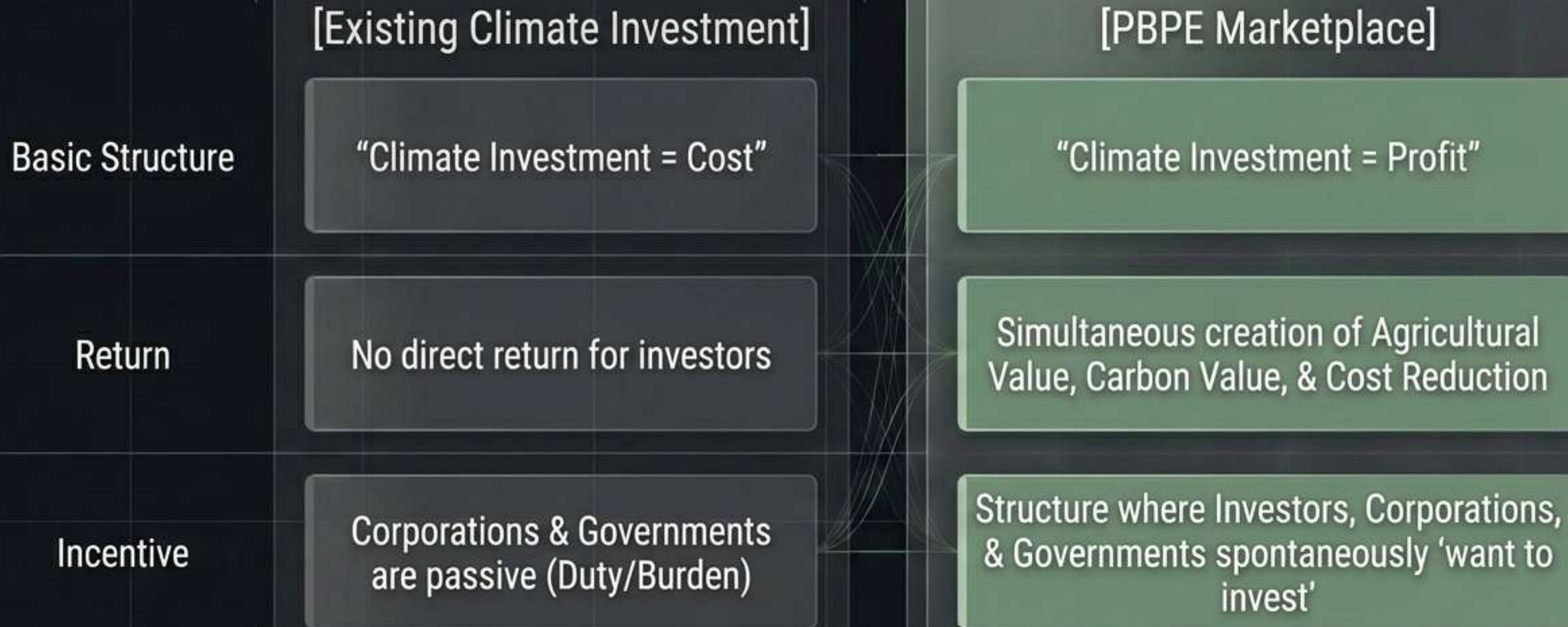
SYSTEM ERROR: FUNDING DEFICIT

Why is existing climate investment always "underfunded"?

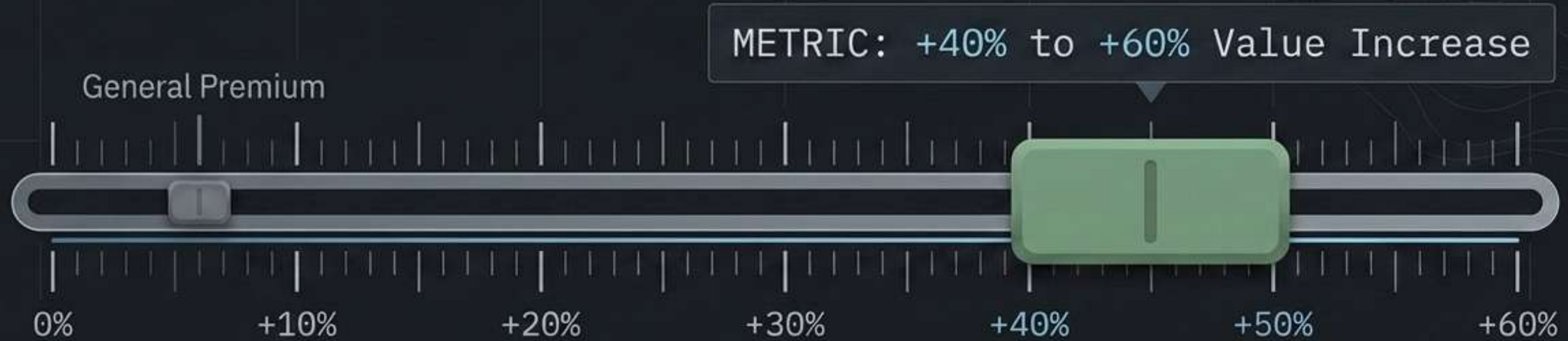
The reason why funds are overwhelmingly insufficient relative to the required investment amount. It is because existing climate investment on "philanthropic funds that disregard how funds" and capital markets as a sustainability and market factor.

1. Low profitability (invests return for direct returns)
2. Slow return on investment
3. Both companies and governments are treated as a mere cost for ESG by not the governments

Structural Reversal by PBPE Marketplace



Changing the Dimension of Green Premium



Definition

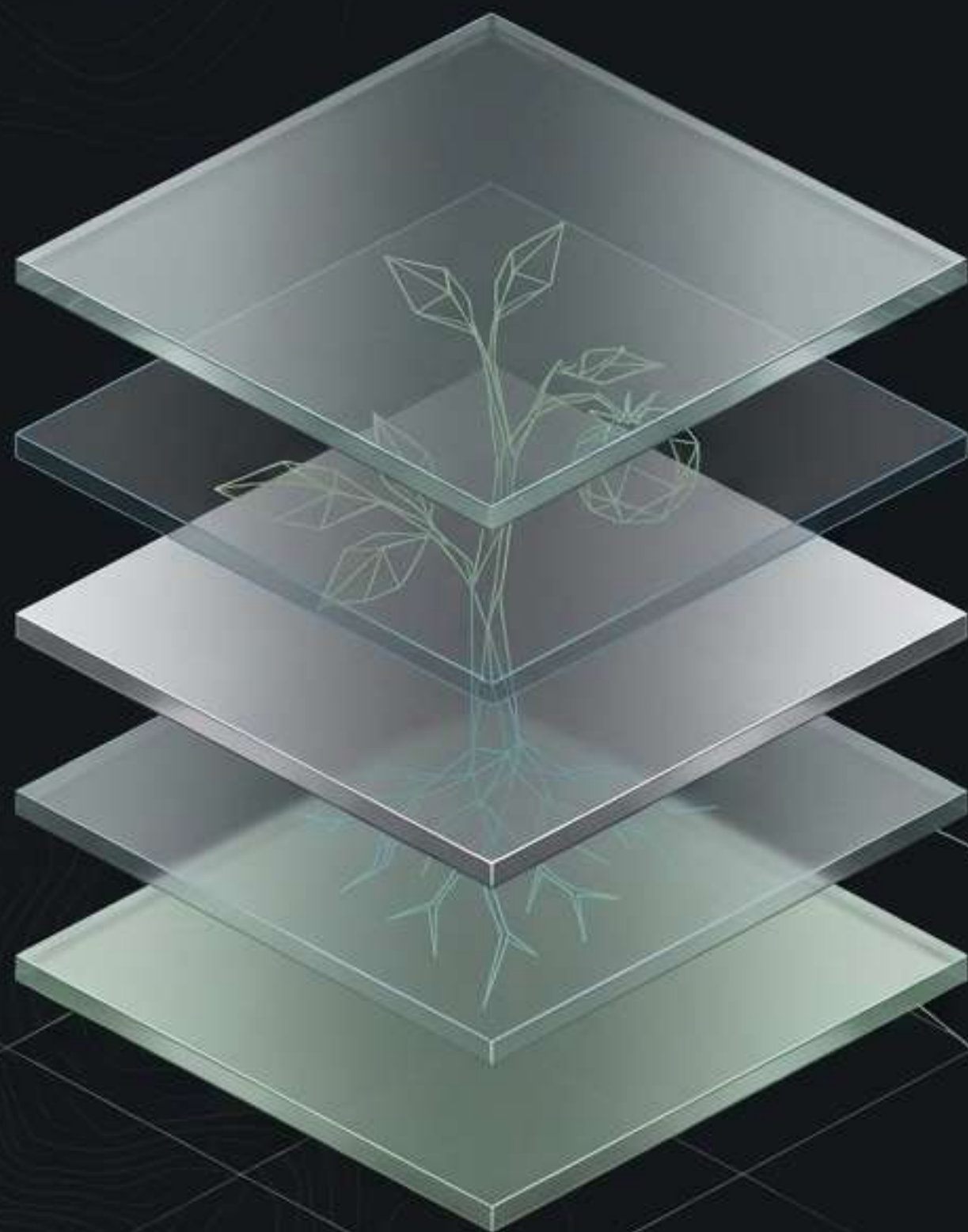
Not the general “additional price due to environmental consideration”, but an overwhelming additional value of agricultural products with “ecosystem, health, and circular value” based on real demand, independent of subsidies or donations.

Case Example & Sources

【Nairobi Model Achievement】
Conventional Tomato: ¥100 → MBT Tomato: ¥140 ~ ¥160
Components of this difference:

- Consumer's 'Health Value'
- Retailer's 'Loss Reduction Value'
- Corporate 'ESG Value'
- PBPE's 'Carbon Value'

Green Premium 5-Layer Structure Complete Model



GREEN PREMIUM = Total +40% ~ +60%

⑤ Regional Value Premium (+5~10%)

Improved Farmer Income • Regional Economic Circulation • Smallholder Farmer Support

④ ESG Premium (+10~20%)

Scope 3 Reduction • Carbon Reduction via PBPE • Water & Soil Improvement

③ Health Premium (+5~15%)

Antioxidant Capacity • Intestinal Environment Improvement • Medical Cost Reduction Potential

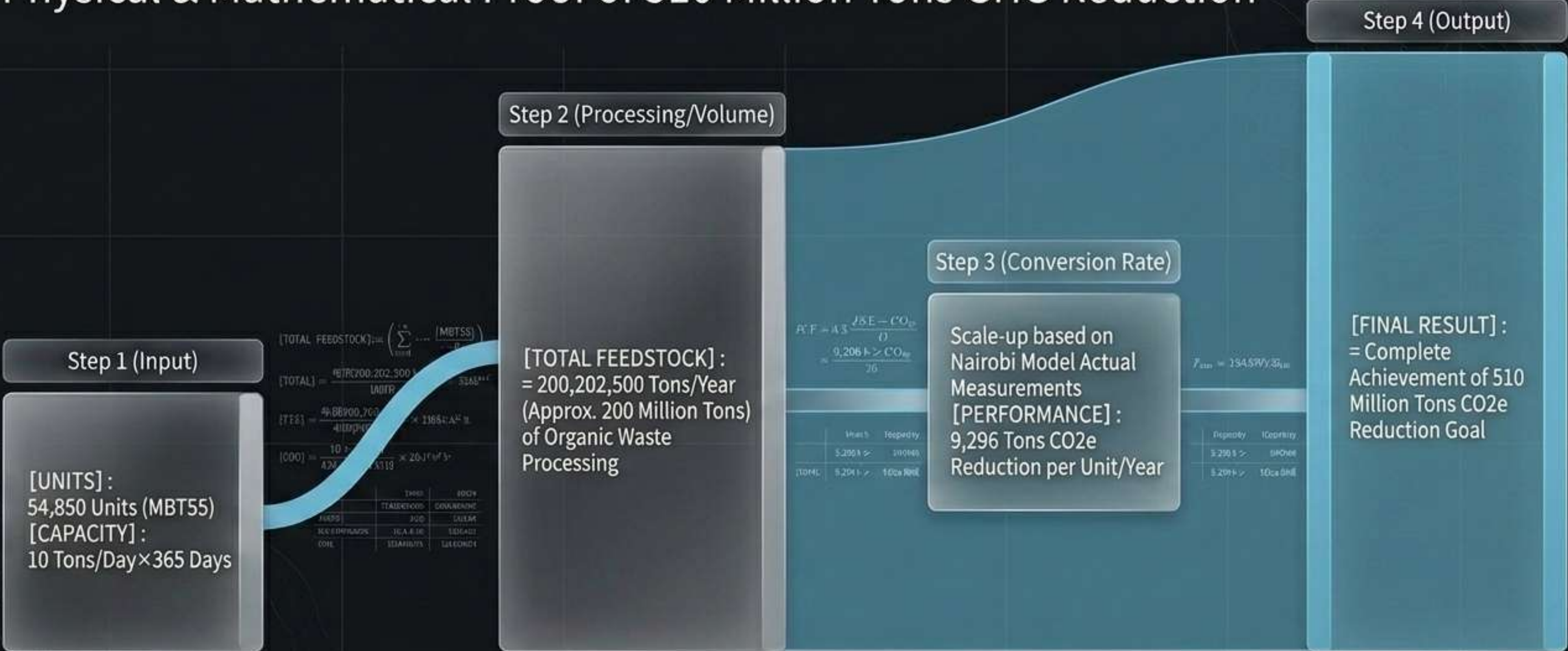
② Preservation Premium (+5~10%)

Freshness Preservation via MBTSS/HMT/NASARA • Waste Loss Reduction • Extended Distribution Distance

① Quality Premium (+10~20%)

Sugar Content • Aroma • Texture • Nutritional Value (Polyphenols • Minerals)

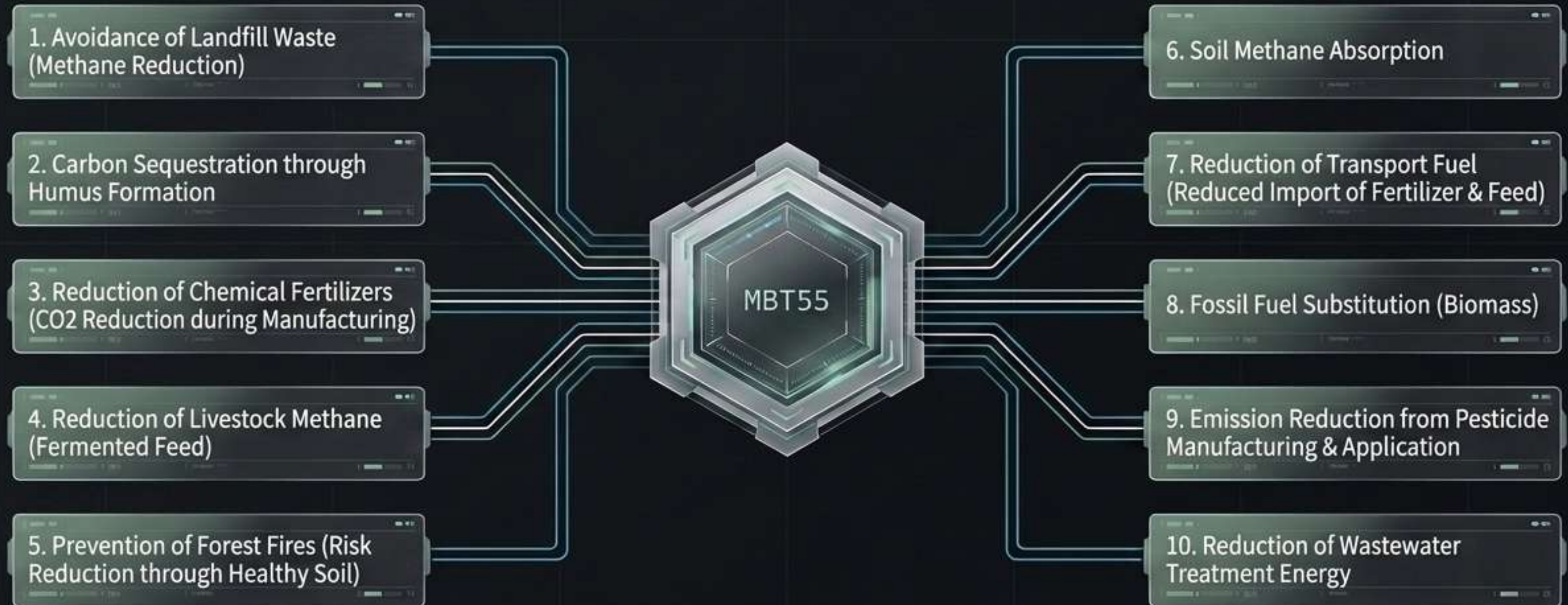
Physical & Mathematical Proof of 510 Million Tons GHG Reduction



Scientific achievement based on empirical data through the physical process of humification → carbon sequestration → methane suppression.

Complete Integration of 10 Reduction Sources

The 510 million ton reduction is not a single reduction method, but the total value of all 10 items integrating all functions of MBT55, far surpassing any single approach.



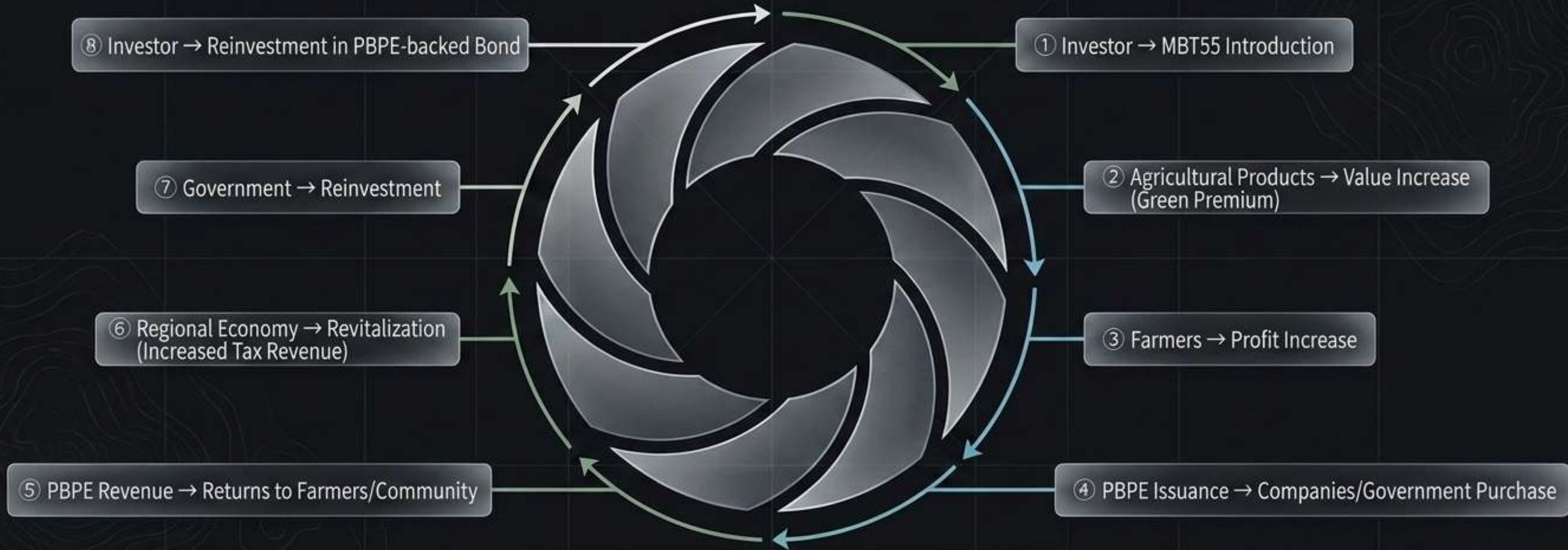
Nairobi Benchmark: Green Premium × PBPE Integrated ROI Model

SCALE: 54,850 Units



“Invest because it’s profitable.” “Participate because ESG is profitable.”
— A groundbreaking economic rationality that completely resolves the funding gap for climate investment.

PBPE Marketplace Capital Circulation System



STATUS:
SUSTAINABLE CIRCULAR
ECONOMY ESTABLISHED

Global Consensus: Full Alignment with World Bank Standards

WORLD BANK NEEDS

Climate finance for agriculture and food systems is currently only "4.3% (approx. \$9 billion)".

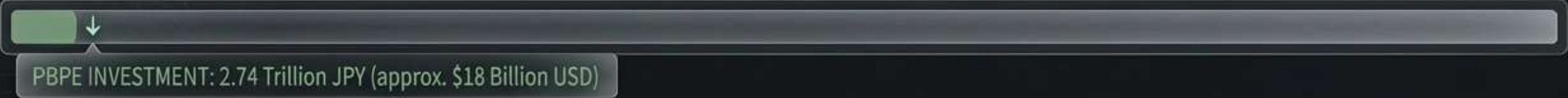
Investment priority areas: Soil, Microorganisms, Water, Smallholder Farmers, Food System Transformation.

PBPE OUTPUT

Embodiment of Nature-based Solutions and Climate-Smart Agriculture.

Perfectly aligned, positioned at the core of all priority areas.

REQUIRED SCALE: 1.2–1.7 Trillion USD per year (World Bank's Required Climate Investment Frame)



Insight: PBPE investment scale is only "1.5% or less" of the World Bank's climate investment frame. An extremely realistic and actionable international solution.



"From Cost to Profit" – The New OS for Planetary Health

VALUE

The only model turning climate investment into 'profit'
(Green Premium +40-60%)

IMPACT

Physical demonstration achieving
510 million tons of GHG reduction
from 200 million tons of waste

ECONOMY

ROI 344%, payback 3.5 months.
Construction of a 2.74 trillion JPY
scale ultra-efficient circular economy

System architecture proof complete.
NEXT INITIATION: Global "PBPE-backed Bond" financial implementation.